ATLAS: Anytime-valid Testing of Latent Simulators Sequential Certification of World-Model Faithfulness via E-Process Betting

Abstract

A world model—a learned, action-conditioned simulator of an environment—is useful only insofar as its imagined rollouts remain *faithful* to reality. Yet faithfulness is a moving target: it decays with the rollout horizon, because recursive prediction compounds one-step error, and it drifts over deployment time, because of non-stationarity, the sim-to-real gap, and the self-referential feedback loop of model-based reinforcement learning, in which an improving policy continually pushes the agent into regions the model never learned. The prevailing practice—scoring a world model once, offline, with metrics such as Fréchet Video Distance, PSNR, or held-out return gaps—is structurally incapable of tracking this degradation. We introduce ATLAS, a framework that equips a deployed world model with a running, statistically valid certificate of the form "as of now, this model's rollouts are trustworthy up to horizon $h^\star(t)$ at tolerance ε," with a guarantee that holds uniformly over all time. ATLAS scores each horizon with a strictly proper scoring rule—which makes it agnostic to the model's representation, covering pixel, latent, and embedding predictors alike—and tests the null hypothesis that the horizon is ε-faithful by betting against it with a test martingale. Combining the resulting per-horizon e-processes yields the *faithfulness frontier* $h^\star(t)$: a time-evolving trust horizon that a planner may legally use to clip its imagination depth, turning a passive monitor into an active controller. The construction is conditional—it places one bet at a time against a conditional-mean null—so it never invokes independence or exchangeability and remains valid on the endogenous, policy-dependent data stream that breaks conformal methods. We establish and prove three guarantees: a Ville-type theorem that a genuinely faithful horizon is revoked with probability at most α, uniformly over all rounds; an anytime-valid simulation lemma that converts the running, empirically certified one-step error into a time-uniform bound on the h-step value gap; and an order-optimal bound on the delay to detect a faithfulness breach. Empirically, we validate each theorem on a controllable latent simulator where ground truth is known, demonstrate ATLAS on real image sequences under three qualitatively different distribution shifts, and monitor learned neural world models trained on a GPU. In every setting the trust horizon collapses precisely when reality diverges from imagination, while offline metrics remain blind.

Keywords: world models, anytime-valid inference, e-processes, sequential testing, model-based reinforcement learning, uncertainty quantification, change detection

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1. Introduction

Consider a model-based agent midway through deployment. It has learned a world model of its environment and uses it constantly: at each step it imagines several possible futures, dozens of steps deep, and commits to the plan that looks best. For a while the imagined futures are excellent—the model was trained on exactly this kind of situation. But the world has quietly changed. A joint has worn, friction has increased, a payload has shifted; or the agent's own improving policy has carried it into a corner of the state space it has never visited before. The world model, oblivious, keeps producing confident rollouts, and the agent keeps trusting them. Nothing in its monitoring stack raises an alarm, because the only number anyone computed—an offline score on a held-out set, measured before deployment began—has not moved. The failure is silent, and by the time it manifests as a bad decision, the certificate that was supposed to guard against it has long since gone stale.

This scenario is the default, not the exception. World models—action-conditioned generative or predictive models of environment dynamics—have become central to modern sequential decision making, from latent-imagination agents [1, 2] to joint-embedding predictive architectures [3, 4] and large-scale video simulators. Their entire value proposition rests on *faithfulness*: an imagined h-step future is useful only if it matches what reality would have produced. Yet the tools used to certify faithfulness are fundamentally mismatched to how faithfulness actually behaves.

Two properties make world-model faithfulness a genuinely sequential, multi-scale quantity. First, it *decays with the horizon*: because a rollout applies the model recursively, a small per-step error compounds, so a model that is essentially perfect one step ahead can be worthless twenty steps ahead. Faithfulness is therefore not a scalar but a profile over horizons. Second, it *drifts over deployment time*: environments are non-stationary, there is an ever-present sim-to-real gap, and—most insidiously—model- based reinforcement learning closes a feedback loop in which the policy that generates the data is itself optimized *through* the model under test. As the policy improves it is driven toward states where the model predicts high value; if the model is optimistically wrong there, the agent is steered precisely into its blind spots. The data distribution is thus *endogenous*: it is a function of the very object we wish to audit.

The dominant evaluation metrics—Fréchet Video Distance [5], pixel-space reconstruction scores, and return gaps—are computed once, on a fixed dataset, independent of the live deployment. By construction they cannot react to horizon-specific decay or to temporal drift; a frozen number is blind to a moving target. One might hope to reach for conformal prediction [6], the modern workhorse of distribution-free uncertainty quantification. But conformal methods and their distribution-shift variants [7, 8, 9] are built for (approximately) exchangeable or bounded-drift single-step prediction; they deliver marginal coverage rather than an anytime-valid, per-horizon certificate, and, decisively, they do not accommodate a data stream whose generating law is a function of the model being tested. Recursive multi-step rollouts and policy endogeneity are exactly the regimes in which the exchangeability crutch snaps.

What the situation demands is a *running certificate*: at every round t, a statement of the form "as of now, this world model's rollouts are trustworthy up to horizon $h^\star(t)$ at error level ε," that is (a) anytime-valid, so an analyst may look at it continuously and act the moment it changes; (b) horizon-resolved, reporting how deep one may trust rather than a single scalar; (c) representation-agnostic, working whether the model predicts pixels, latent states, or embeddings; and (d) sound under endogeneity, remaining valid when the policy generating the data depends on the model. This is precisely the profile of an *e-process* [10, 11], the sequential-testing object whose defining feature is time-uniform validity under optional stopping and continuation. Offline metrics cannot provide it; e-processes can.

We propose **ATLAS** (Anytime-valid Testing of Latent Simulators). At each round the model emits, for each horizon h, an action-conditioned predictive distribution; reality later reveals the realized outcome. ATLAS scores the prediction against the outcome with a strictly proper scoring rule [12] and bets against the null hypothesis that the model is ε-faithful at horizon h, accumulating wealth through a test martingale whenever the evidence contradicts the null [13, 14, 15]. It runs one such e-process per horizon and reports the faithfulness frontier

$$h^\star(t) = \text{the largest horizon all of whose e-processes have yet to reject,}$$

a time-evolving trust horizon. Because $h^\star(t)$ carries an anytime-valid guarantee, a planner may clip its rollout depth to $h^\star(t)$ at every step and inherit that guarantee, so ATLAS is not merely a monitor but a controller. The keystone is that every bet is placed *conditionally*, requiring only that the expected wealth increment be at most one given the past; this constrains a single conditional expectation, so it holds for any predictable policy and betting rule, and the endogeneity that defeats exchangeability-based methods is handled automatically.

Figure 1 previews the central phenomenon.

1.1. Contributions

1. **A conditional e-process construction for world-model faithfulness on endogenous streams** (Sections 2–3). We formalize ε-faithfulness as a horizon-indexed, conditional-mean null under the deployment filtration and build per-horizon test martingales by betting, using mixture (growth-rate-optimal) bets. We identify and resolve the recursive-multi-step validity problem that invalidates naive multi-step conformal prediction.

2. **The faithfulness frontier** (Section 3): a horizon-indexed family of dependent e-processes, a valid aggregation across horizons under arbitrary dependence, and a monotone trust-horizon estimator $h^\star(t)$ that doubles as a planning controller.

3. **Three theorems, proved in full** (Section 4): uniform validity (a faithful horizon is never revoked except with probability at most α); an anytime-valid simulation lemma that lifts a running, empirically certified one-step error to a time-uniform bound on the h-step value gap—to our knowledge the first time-uniform version of the simulation lemma; and an order-optimal detection-delay bound under a drift alternative.

4. **A three-tier empirical study** (Section 5): a ground-truth validation of all three theorems on a controllable latent simulator; real-data demonstrations on Moving MNIST and the KTH human-action dataset under speed,

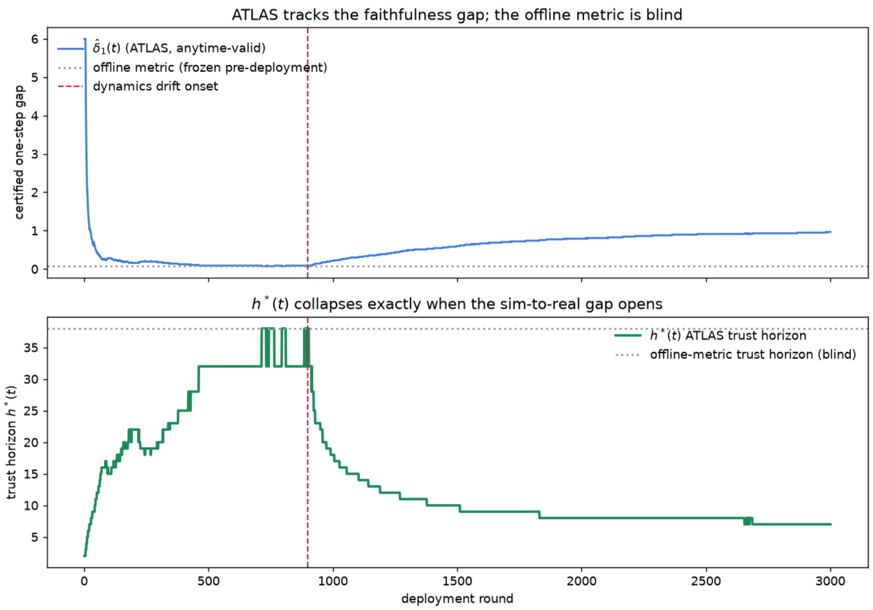


Figure 1: **The central phenomenon.** A world model is faithful; a dynamics shift is injected mid-deployment (dashed line). *Top:* ATLAS's certified one-step faithfulness gap $\hat{\delta}_1(t)$ (an anytime-valid upper confidence sequence) stays low while the model is faithful and rises exactly when the shift opens, whereas an offline metric frozen at its pre-deployment value is blind. *Bottom:* the ATLAS trust horizon $h^\star(t)$ holds high while the model is faithful and collapses within a few rounds of the shift, while the offline-metric trust horizon remains dangerously optimistic. This work makes the collapse a theorem, not merely a curve.

gait, and domain shifts; and ATLAS monitoring learned neural world models—a convolutional autoencoder and a JEPA-style frozen-encoder model—on a GPU. The trust horizon collapses when the shift opens; offline metrics do not react.

1.2. Related work

ATLAS sits at the confluence of four research programs.

Anytime-valid inference, e-values, and testing by betting. The foundation is Ville's inequality [16], which controls the probability that a nonnegative supermartingale ever crosses a high threshold, and the modern theory of game-theoretic probability, e-values, and e-processes [11, 17, 10]. Testing by betting [13] reframes a test as a gambler wagering against the null; safe testing and growth-rate-optimal betting [14] characterize wagers that maximize the rate of evidence accumulation; and betting confidence sequences for bounded means [15], with time-uniform empirical-Bernstein bounds [18], give the tight, variance-adaptive intervals we use for the one-step gap. ATLAS specializes this machinery to world-model faithfulness on action-conditioned, policy-endogenous streams, organized into a horizon-indexed frontier; this combination is new.

Conformal prediction and its time-series variants. Conformal prediction [6] provides distribution-free coverage under exchangeability; covariate-shift [7] and adaptive or online variants [8, 9] relax exchangeability to bounded or adversarial drift. These methods target marginal coverage rather than an anytime-valid per-horizon guarantee, degrade under recursive multi-step rollouts, and presume an exogenous data-generating mechanism. ATLAS's conditional construction abandons exchangeability altogether, which is what lets it handle multi-step, endogenous streams.

World models and their evaluation. World models span latent-imagination agents [1, 2] and joint-embedding predictive architectures [3, 4], but their evaluation has remained offline: Fréchet Video Distance [5], reconstruction metrics, and return gaps all report a single static number. ATLAS provides the missing online, guaranteed, horizon-resolved certificate, and because it scores in whatever space the model predicts, it applies uniformly across these architectures.

Model-based reinforcement learning and the simulation lemma. The simulation lemma [19] bounds the difference in value between two Markov decision processes by their average one-step dynamics discrepancy, underlying model-based policy optimization [20] and recent tightness analyses [21]. All existing forms are fixed-sample and assume the one-step error is known. Our simulation lemma (Theorem 2) is, to our knowledge, the first anytime-valid version, driven by a running confidence sequence and holding uniformly over deployment time.

Sequential change detection. Classical change detection and the Lorden/Lai minimax delay theory [22, 23] address detecting a shift with minimal delay under a false-alarm constraint; the recent e-detector framework [24] recasts these as sums of e-processes with nonparametric guarantees. We specialize e-detectors to a score-divergence faithfulness null and couple their delay to the collapse of the trust horizon.

The nearest anytime-valid neighbors monitor agent *behavior* or attribute *evaluation* drift in language-model pipelines; ATLAS instead audits what an agent's world model *believes*, complementary and, to our knowledge, unoccupied ground.

2. Problem formulation

2.1. Deployment protocol and notation

A world model M is deployed online alongside a policy or planner π. Deployment proceeds in rounds $t = 1, 2, \ldots$. At round t the agent occupies a (possibly latent) state $x_t \in \mathcal{X}$ and emits an observation; the planner selects an action sequence $a_{t:t+H-1} = (a_t, \ldots, a_{t+H-1})$ up to a maximum horizon H, using M for imagined rollouts. For each horizon $h \in \{1, \ldots, H\}$ the model produces an action-conditioned h-step predictive distribution

$$P^M_{t,h}(\cdot) := P^M\big(\cdot \mid x_t,\, a_{t:t+h-1}\big) \in \Delta(\mathcal{Y}), \tag{1}$$

over a target space $\mathcal{Y}$ (observations, states, or a learned latent code). The agent executes a prefix of the plan; reality reveals the realized h-step outcome $Y_t^{(h)} \in \mathcal{Y}$, which resolves the round-t, horizon-h prediction at time $t+h$. We index each per-horizon stream by resolution order and write k for the k-th resolved horizon-h prediction, issued at time s_k and resolved at s_k+h. We work on a filtered probability space with the deployment filtration

$$\mathcal{F}_k := \sigma\big(\text{all states, actions, planner internals, and outcomes observable up to the resolution of the} \tag{2}$$

and make no assumption that the data are independent or exchangeable. A random variable is *predictable* if it is $\mathcal{F}_{k-1}$-measurable.

2.2. Scoring rules and score divergence

Fix a negatively oriented, strictly proper scoring rule $S : \Delta(\mathcal{Y}) \times \mathcal{Y} \to [0, B]$ [12]: smaller is better, and $\mathbb{E}_{Y\sim Q}[S(Q, Y)] \leq \mathbb{E}_{Y\sim Q}[S(P, Y)]$ for all P, Q, with equality iff $P = Q$. The associated score divergence is

$$d_S(Q, P) := \mathbb{E}_{Y\sim Q}[S(P, Y)] - \mathbb{E}_{Y\sim Q}[S(Q, Y)] \;\geq\; 0, \qquad d_S(Q, P) = 0 \iff P = Q. \tag{3}$$

Boundedness is used only for the empirical-Bernstein rates and can be relaxed to a conditional sub-ψ tail condition. Three instances recur.

Example 1 (Logarithmic score / KL)**.** For $S(P, y) = -\log p(y)$, $d_S(Q, P) = \mathrm{KL}(Q \,\|\, P)$; by Pinsker's inequality $\mathrm{TV}(Q, P) \leq \sqrt{\frac{1}{2}\mathrm{KL}(Q\|P)}$, giving the score-to-metric modulus $\phi(u) = \sqrt{u/2}$ with $\rho = \mathrm{TV}$.

Example 2 (Energy score / MMD)**.** For the energy score, d_S is a power of the energy distance, an integral-probability metric; the modulus ϕ is the corresponding one. This is the natural choice for pixel and sample-based predictors.

Example 3 (Gaussian / Mahalanobis score)**.** For a Gaussian predictive $\mathcal{N}(\mu, \Sigma)$, the log score's divergence controls a squared Wasserstein-2 distance, and the squared Mahalanobis distance $(y-\mu)^\top \Sigma^{-1}(y-\mu)$ is χ^2_d-distributed under calibration. This makes ATLAS applicable to latent predictors *without decoding to pixels*: one scores the predicted embedding against the realized embedding in the model's own latent metric. Such representation-agnosticism is what lets a single method span pixel, latent, and embedding world models.

2.3. The faithfulness null

Let $Q_{s,h} := \mathrm{Law}(Y_s^{(h)} \mid \mathcal{F}_s)$ be the true conditional law of the h-step outcome given the information at issue time.

Definition 1 (ε-faithfulness)**.** For horizon h and tolerance $\varepsilon_h \geq 0$, the model is ε_h-faithful at horizon h if $H_0^{(h)}:\ d_S(Q_{s,h}, P^M_{s,h}) \leq \varepsilon_h$ for every issue-time s.

$\varepsilon_h = 0$ is exact conditional calibration; $\varepsilon_h > 0$ tolerates a fixed, application-chosen amount of model error. In practice we operationalize $H_0^{(h)}$ through a per-round statistic Z_k whose conditional mean is controlled by ε_h under the null (Section 3); two canonical constructions are the excess of the model's score over a predictable calibrated reference, and the calibration excess of a latent Gaussian predictive.

2.4. The endogeneity obstruction

The defining difficulty of the world-model setting is that the data stream is policy-dependent: the distribution of $x_{t+1}, x_{t+2}, \ldots$ depends on the actions the planner chose by imagining rollouts inside M. Two consequences follow. First, the outcomes $\{Y_t^{(h)}\}$ are neither independent nor exchangeable; any method whose validity rests on exchangeability loses its guarantee here. Second, and more subtly, the data-generating mechanism is a function of the object under test: as π is optimized through M, it is driven toward states where M predicts high value, so if M is optimistically miscalibrated there, the test distribution concentrates on the model's worst regions and a fixed reference distribution moves against the tester.

ATLAS resolves both by never invoking exchangeability. It requires only that each betting increment e_k satisfy the conditional inequality $\mathbb{E}[e_k \mid \mathcal{F}_{k-1}] \leq 1$ under the null. Because this constrains a single conditional expectation given the entire past—including all prior policy decisions and the current context—it holds for any predictable policy and any predictable betting rule. Endogeneity is thus not an obstacle to be modeled but a regime the conditional construction handles automatically.

3. The ATLAS method

3.1. E-values, test supermartingales, and e-processes

An *e-variable* for a null H_0 is a nonnegative random variable e with $\mathbb{E}_{H_0}[e] \leq 1$; large values are evidence against H_0. A *test supermartingale* is a process $W_0 = 1$, $W_k = \prod_{j \leq k} e_j$, in which each e_j is a conditional e-variable, $\mathbb{E}[e_j \mid \mathcal{F}_{j-1}] \leq 1$ under H_0; it is a nonnegative supermartingale under H_0. An *e-process* is a nonnegative process dominated by a test supermartingale under every law in a composite null. The engine of anytime validity is the following classical inequality.

Lemma 1 (Ville's inequality [16])**.** *If $(W_k)_{k \geq 0}$ is a nonnegative supermartingale with $W_0 = 1$, then for any $\alpha \in (0,1)$, $\mathbb{P}(\sup_{k \geq 0} W_k \geq 1/\alpha) \leq \alpha$.*

3.2. The per-horizon betting e-process

Fix a horizon h. Let $Z_k \in [z_{\text{lo}}, z_{\text{hi}}]$ be an $\mathcal{F}_k$-measurable per-round statistic—the score advantage—designed so that the null controls its conditional mean:

$$H_0^{(h)} \implies \mathbb{E}\big[Z_k \mid \mathcal{F}_{k-1}\big] \;\leq\; \varepsilon_h \qquad \text{for all } k. \tag{4}$$

For instance, with a predictable calibrated reference predictor R_k, taking $Z_k = S(P^M_{s_k,h}, Y^{(h)}_{s_k}) - S(R_k, Y^{(h)}_{s_k})$ makes (4) the statement that the model does not lose to the reference by more than ε_h; with a latent Gaussian predictive, taking $Z_k = \text{mahalanobis}^2/d - 1$ makes $\mathbb{E}[Z_k \mid \mathcal{F}_{k-1}] = 0$ under exact calibration, so (4) holds with $\varepsilon_h = 0$.

Given a predictable betting fraction $\lambda_k \in [0, \lambda_{\max}]$ with $\lambda_{\max} = 1/(z_{\text{hi}} - z_{\text{lo}})$, define the betting e-variable and wealth

$$e_k := 1 + \lambda_k\,(Z_k - \varepsilon_h), \qquad W_K^{(h)} := \prod_{k=1}^{K} e_k. \tag{5}$$

Proposition 1 (Per-round validity)**.** *Assume* (4) *and* $\varepsilon_h \in [z_{\mathrm{lo}}, z_{\mathrm{hi}}]$*. Then for every* k*,* $e_k \geq 0$ *and* $\mathbb{E}[e_k \mid \mathcal{F}_{k-1}] \leq 1$*; consequently* $(W_K^{(h)})_K$ *is a nonnegative test supermartingale under* $H_0^{(h)}$*.*

Proof. Since $\lambda_k \geq 0$, e_k is minimized over $Z_k \in [z_{\mathrm{lo}}, z_{\mathrm{hi}}]$ at $Z_k = z_{\mathrm{lo}}$, so $e_k \geq 1+\lambda_k(z_{\mathrm{lo}}-\varepsilon_h) \geq 1-\lambda_{\max}(\varepsilon_h - z_{\mathrm{lo}}) \geq 0$, using $\varepsilon_h - z_{\mathrm{lo}} \leq z_{\mathrm{hi}} - z_{\mathrm{lo}} = 1/\lambda_{\max}$. As λ_k is predictable, $\mathbb{E}[e_k \mid \mathcal{F}_{k-1}] = 1 + \lambda_k(\mathbb{E}[Z_k \mid \mathcal{F}_{k-1}] - \varepsilon_h) \leq 1$ by (4). Hence $\mathbb{E}[W_k^{(h)} \mid \mathcal{F}_{k-1}] = W_{k-1}^{(h)} \mathbb{E}[e_k \mid \mathcal{F}_{k-1}] \leq W_{k-1}^{(h)}$ with $W_0^{(h)} = 1$. □

The bet λ_k governs power, not validity. A single well-chosen λ is optimal only if the post-null margin is known; since it is not, we use *mixture betting*: fix a grid $\{\lambda^{(1)}, \dots, \lambda^{(m)}\} \subset (0, \lambda_{\max}]$ and a prior ν, and set $W_K^{(h)} = \sum_j \nu_j \prod_k (1 + \lambda^{(j)}(Z_k - \varepsilon_h))$. Each summand is a test supermartingale by Proposition 1, and a convex combination of test supermartingales is a test supermartingale (Lemma 2); the mixture is parameter-free, does not deplete wealth under the null, and adapts to an unknown margin, realizing the growth-rate-optimal principle of Grünwald et al. [14].

Lemma 2 (Mixtures preserve validity)**.** *If* $(W_k^{(j)})_k$ *are nonnegative test supermartingales and* $\nu_j \geq 0, \sum_j \nu_j = 1$*, then* $\bar{W}_k := \sum_j \nu_j W_k^{(j)}$ *is a nonnegative test supermartingale.*

Proof. By linearity, $\mathbb{E}[\bar{W}_k \mid \mathcal{F}_{k-1}] = \sum_j \nu_j \mathbb{E}[W_k^{(j)} \mid \mathcal{F}_{k-1}] \leq \sum_j \nu_j W_{k-1}^{(j)} = \bar{W}_{k-1}$, and $\bar{W}_0 = 1$. □

3.3. Recursive-multi-step validity

A subtlety specific to multi-step prediction must be addressed for (4) to hold. If, for horizon h, we form a prediction at *every* issue time, then consecutive residuals overlap: the horizon-h residual issued at s and the one issued at $s+1$ share the innovations in the window $(s+1, \dots, s+h-1)$. The residual stream is thus a moving-average process of order $h-1$, and conditioning on the past reveals part of the current residual, so $\mathbb{E}[Z_k \mid \mathcal{F}_{k-1}] \neq \varepsilon_h$ in general even when the model is faithful—precisely why naive multi-step conformal prediction loses validity. We restore (4) by processing each horizon on *non-overlapping* windows (issue times $s \in \{0, h, 2h, \dots\}$), so that residuals are conditionally independent across the per-horizon index. The cost is real—horizon h receives a factor h fewer updates—and honestly encodes that long horizons are intrinsically harder to certify.

Remark 1 (Recovering data efficiency)**.** Partitioning issue times into h offset classes yields h non-overlapping chains, each valid on its own coarse filtration by Proposition 1. Rejecting horizon h when *any* class crosses Hh/α uses every issue time while retaining level-α validity, at an additional $\log h$ in the threshold; this recovers the data efficiency lost to non-overlap at a logarithmic price.

3.4. The faithfulness frontier

Run one e-process $W^{(h)}$ per horizon and declare horizon h *revoked* at the first time its wealth crosses the threshold, $\tau_h := \inf\{k : W_k^{(h)} \geq 1/\alpha\}$. Revocation is permanent: since Ville's inequality concerns the event that the wealth *ever* crosses $1/\alpha$, we track the running maximum and never un-revoke. The faithfulness frontier is the monotone trust horizon

$$h^\star(t) := \max\{\, h : \text{no horizon } h' \leq h \text{ has been revoked by round } t\,\}, \qquad (6)$$

with $h^\star(t) = 0$ if horizon 1 is revoked. Monotonicity encodes the physical fact that a broken short horizon implies broken longer ones, and it matches the intended downstream use, *clipping imagination depth*: a planner sets its rollout depth to $\min(H_{\text{desired}}, h^\star(t))$ and, by Theorem 1, plans on an untrustworthy horizon only with probability at most α over the entire deployment.

For simultaneous control of the whole frontier we either run horizon h at level α/H (a union bound, costing $+\log H$) or, when a single pooled certificate suffices, drive the frontier off the averaged e-process $\bar{W}_k = \frac{1}{H}\sum_h W_k^{(h)}$, which by Lemma 2 is valid under arbitrary dependence across horizons—essential, since the horizon streams share the same underlying trajectory. When the object is to detect a *change* in faithfulness rather than to test a fixed horizon, we use a Shiryaev–Roberts e-detector [24], $R_t = (1 + R_{t-1})e_t$, which accumulates evidence over all candidate changepoints so that power is not depleted before the change. Algorithm 1 summarizes the procedure.

4. Theoretical guarantees

Assumption 1 (Bounded score)**.** $S \in [0, B]$, hence $Z_k \in [z_{\text{lo}}, z_{\text{hi}}]$ with $z_{\text{hi}} - z_{\text{lo}} \leq B'$.

Assumption 2 (Predictability)**.** The bets λ_k and any reference R_k are $\mathcal{F}_{k-1}$-measurable; no independence or exchangeability is assumed.

Algorithm 1 ATLAS: anytime-valid faithfulness certification

input: horizons $\{1,\dots,H\}$, tolerances $\{\varepsilon_h\}$, level α, bet grid $\{\lambda^{(j)}\}$ and prior ν, proper score S
initialize wealth $W^{(h)} \leftarrow 1$ and `revoked`$(h) \leftarrow$ false for all h
for each deployment round $t = 1, 2, \dots$ **do**
    for each horizon h resolving a non-overlapping window at t **do**
        observe outcome $Y^{(h)}$; form the score statistic $Z \in [z_{\mathrm{lo}}, z_{\mathrm{hi}}]$
        $W^{(h)} \leftarrow \sum_j \nu_j \, \Pi_j \, (1+\lambda^{(j)}(Z-\varepsilon_h))$, updating each running product Π_j
        if $W^{(h)} \geq H/\alpha$ **then** set `revoked`$(h) \leftarrow$ true
        end if
    end for
    $h^\star(t) \leftarrow \max\{h : \neg$`revoked`$(h')$ for all $h' \leq h\}$
    output the trust horizon $h^\star(t)$; the planner clips its rollout depth to $h^\star(t)$
end for

Assumption 3 (Score-to-metric modulus)**.** There is a nondecreasing, concave ϕ with $\rho(Q,P) \leq \phi(d_S(Q,P))$ for a probability metric ρ (e.g. Pinsker: $\rho = \mathrm{TV}$, $\phi(u) = \sqrt{u/2}$).

Assumption 4 (Rollout regularity)**.** Either (i) rewards lie in $[0, R_{\max}]$ and returns are γ-discounted, with a bounded concentrability coefficient C_∞ relating the rollout-visited state–action distribution to the deployment distribution on which the one-step gap is certified; or (ii) the true transition kernel is L-Lipschitz in ρ.

4.1. Uniform validity: no false revocation

Theorem 1 (Uniform validity)**.** *Fix a horizon h and $\alpha \in (0,1)$. Under $H_0^{(h)}$ and Assumptions 1–2, $\mathbb{P}(\exists K \geq 1 : W_K^{(h)} \geq 1/\alpha) \leq \alpha$; hence the probability that ATLAS ever revokes a truly ε_h-faithful horizon is at most α, uniformly over all rounds and over all predictable policies and betting rules. Running each of the H horizons at level α/H gives simultaneous control of the entire frontier: writing $\mathcal{H}_0$ for the set of genuinely faithful horizons, $\mathbb{P}(\exists t,\ \exists h \in \mathcal{H}_0 :\ h$ revoked by round $t) \leq \alpha$.*

Proof. By Proposition 1 and Lemma 2, $(W_K^{(h)})_K$ is a nonnegative supermartingale with $W_0^{(h)} = 1$ under $H_0^{(h)}$; this uses only the conditional-mean null (4) and predictability, never exchangeability. Ville's inequality gives $\mathbb{P}(\sup_K W_K^{(h)} \geq 1/\alpha) \leq \alpha$, and the revocation event $\{\tau_h < \infty\}$ equals $\{\sup_K W_K^{(h)} \geq 1/\alpha\}$. For the frontier, the event that some faithful horizon is ever revoked lies in $\bigcup_{h \in \mathcal{H}_0} \{\sup_K W_K^{(h)} \geq H/\alpha\}$; each has probability at most α/H by Ville at level α/H, and a union bound over $|\mathcal{H}_0| \leq H$ horizons yields α. □

Remark 2 (Composite null and conservativeness)**.** $H_0^{(h)}$ is composite: it contains every conditional law with $d_S \leq \varepsilon_h$, and validity holds because $\mathbb{E}[e_k \mid \mathcal{F}_{k-1}] \leq 1$ is required for every member. The guarantee is one-sided and typically loose—under the null the wealth is a supermartingale and drifts down—so empirical revocation rates are usually far below α (Section 5.1); this is the price, and the safety margin, of anytime validity.

4.2. An anytime-valid simulation lemma

We convert the running, certified one-step error into a time-uniform bound on the h-step value gap. The first step reads Theorem 1 as a confidence sequence.

Lemma 3 (One-step gap confidence sequence)**.** *Instantiate the construction at $h = 1$. For a candidate u, let $\widetilde{W}_K(u) = \prod_{k \leq K}(1 + \lambda_k(u)(u - Z_k))$ with predictable $\lambda_k(u) \in [0, \lambda_{\max}]$, let $m_K := \frac{1}{K}\sum_{k \leq K} \mathbb{E}[Z_k \mid \mathcal{F}_{k-1}]$ be the running one-step gap, and $\hat{\delta}_1(K) := \sup\{u : \widetilde{W}_K(u) < 1/\alpha\}$. Under Assumptions 1–2, $\mathbb{P}(\forall K :\ m_K \leq \hat{\delta}_1(K)) \geq 1 - \alpha$, and with mixture bets $\hat{\delta}_1(K) = m_K + O(\sqrt{\hat{\sigma}_K^2 \log(1/\alpha)/K} + B' \log(1/\alpha)/K)$, the variance-adaptive empirical-Bernstein rate.*

Proof. Consider, for each fixed u, the null $H_u^{\geq} :\ \mathbb{E}[Z_k \mid \mathcal{F}_{k-1}] \geq u$ for all k. Applying Proposition 1 with Z_k replaced by $-Z_k$ and ε_h by $-u$ shows that $\widetilde{W}_K(u)$, which grows when the mean lies below u, is a nonnegative test supermartingale under $H_u^{\geq}$; Ville's inequality gives $\mathbb{P}(\sup_K \widetilde{W}_K(u) \geq 1/\alpha \mid H_u^{\geq}) \leq \alpha$. If the true running mean satisfied $m_K > \hat{\delta}_1(K)$ for some K, then at the boundary value $u = m_K$ we would have $\widetilde{W}_K(u) \geq 1/\alpha$ while $H_u^{\geq}$ holds, an event of probability at most α; since $\widetilde{W}_K(u)$ is monotone in u, testing the boundary suffices. Hence $\mathbb{P}(\exists K : m_K > \hat{\delta}_1(K)) \leq \alpha$. The empirical-Bernstein rate for the mixture choice is that of Waudby-Smith and Ramdas [15, Thm. 2]. □

Theorem 2 (Anytime-valid simulation lemma)**.** *Under Assumptions 1–4, with probability at least $1-\alpha$, simultaneously for all rounds t and all horizons $h \le H$, the gap between the imagined h-step return under M and the true h-step return under real dynamics for the deployed policy π obeys*

$$\big| V^{\pi}_{M,h}(t) - V^{\pi}_{\mathrm{real},h}(t) \big| \;\le\; C(h)\cdot\phi\big(\hat{\delta}_1(t)\big), \qquad C(h) = \begin{cases} \dfrac{\gamma(1-\gamma^h)}{(1-\gamma)^2}\, C_\infty R_{\max}, & \textit{case (i)}, \\ R_{\max} \sum_{i=1}^{h} L^{\,i-1}, & \textit{case (ii)}, \end{cases} \tag{7}$$

where $\hat{\delta}_1(t)$ is the certified one-step gap of Lemma 3.

Proof. By Lemma 3, with probability $\ge 1-\alpha$ the bound $\hat{\delta}_1(t)$ upper-bounds the running one-step gap m_t for all t simultaneously. By Assumption 3 and Jensen's inequality (using concavity of ϕ), the average one-step error in ρ satisfies $\mathbb{E}[\rho(Q_{\cdot,1}, P^M_{\cdot,1})] \le \mathbb{E}[\phi(d_S)] \le \phi(\mathbb{E}[d_S]) = \phi(m_t) \le \phi(\hat{\delta}_1(t))$, the last step by monotonicity. To propagate one-step error through the rollout, write the h-step law as the h-fold composition of one-step kernels and telescope, $\mathrm{Law}^M_{t,h} - \mathrm{Law}^{\mathrm{real}}_{t,h} = \sum_{i=1}^{h} (\mathcal{K}^{\mathrm{real}})^{h-i} (\mathcal{K}^M - \mathcal{K}^{\mathrm{real}}) (\mathcal{K}^M)^{i-1}$, where $\mathcal{K}$ is the one-step transition operator. In case (ii), each true kernel is a ρ-contraction of modulus L, so by the triangle inequality $\rho(\mathrm{Law}^M_{t,h}, \mathrm{Law}^{\mathrm{real}}_{t,h}) \le \sum_{i=1}^{h} L^{i-1}\,\bar{\rho}_1$ with $\bar{\rho}_1 \le \phi(\hat{\delta}_1(t))$; multiplying by $R_{\max}$ gives the case-(ii) bound. In case (i), the discounted value-difference telescoping is the simulation lemma of Kearns and Singh [19], $|V^{\pi}_M - V^{\pi}_{\mathrm{real}}| \le \frac{\gamma}{(1-\gamma)^2} R_{\max}\, \mathbb{E}_{d^\pi}[\rho_{\mathrm{TV}}]$, with truncation to horizon h replacing 1 by $1-\gamma^h$; the one-step error is evaluated on the rollout-visited distribution d^π, at most C_∞ times its certified value on the deployment distribution. Substituting $\bar{\rho}_1 \le \phi(\hat{\delta}_1(t))$ yields (7). The union over t and h is free: Lemma 3 holds uniformly in t and the telescoping is deterministic given $\hat{\delta}_1(t)$. □

Remark 3 (Why this is the bridge)**.** Every quantity on the right of (7) is observed and certified online: a practitioner reads off a live, guaranteed upper bound on how far imagined rollouts can drift from reality. The trust horizon $h^\star(t) = \max\{h : C(h)\phi(\hat{\delta}_1(t)) \le \text{budget}\}$ follows directly, and since $C(h)$ is increasing in h, a rise in $\hat{\delta}_1$ collapses the horizon monotonically—the mechanism behind Figure 1.

4.3. Detection delay under a faithfulness breach

Theorem 3 (Detection delay)**.** *Suppose $\mathbb{E}[Z_k \mid \mathcal{F}_{k-1}] \le \varepsilon_1$ for $k \le \tau$ and $\mathbb{E}[Z_k \mid \mathcal{F}_{k-1}] \ge \varepsilon_1 + \Delta$ with conditional variance at most σ^2 for $k > \tau$,* *at an unknown changepoint τ. Run the ATLAS mixture e-process (or the Shiryaev–Roberts e-detector) with alarm time $T = \inf\{t : W_t \geq 1/\alpha\}$. Then* (a) *under the null $\mathbb{P}(T < \infty) \leq \alpha$ for the e-process and $\mathbb{E}_\infty[T] \geq 1/\alpha$ for the e-detector; and* (b) *with probability at least $1 - \alpha$,*

$$T - \tau \;\leq\; \frac{\log(1/\alpha)}{D^\star}\,(1 + o(1)), \qquad D^\star = \sup_{\lambda \in [0, \lambda_{\max}]} \mathbb{E}_{\text{post}}\big[\log(1 + \lambda(Z - \varepsilon_1))\big],$$

with $D^\star \geq \Delta^2/\big(2(\sigma^2 + B'\Delta/3)\big)$ for the bounded score, hence $T - \tau \lesssim 2(\sigma^2 + B'\Delta/3)\log(1/\alpha)/\Delta^2$. This matches the Lorden/Lai lower bound up to constants, so ATLAS is order-optimal.

Proof. (a) The e-process claim is Theorem 1. For the e-detector, $\mathbb{E}[R_t \mid \mathcal{F}_{t-1}] = (1 + R_{t-1})\mathbb{E}[e_t \mid \mathcal{F}_{t-1}] \leq 1 + R_{t-1}$, so $(R_t - t)$ is a supermartingale; optional stopping at T gives $\mathbb{E}_\infty[R_{T\wedge n}] \leq \mathbb{E}[T \wedge n]$, and since $R_T \geq 1/\alpha$ on $\{T < \infty\}$, letting $n \to \infty$ yields $\mathbb{E}_\infty[T] \geq 1/\alpha$. (b) Post-change, the log-wealth increments $\log(1 + \lambda(Z_k - \varepsilon_1))$ are bounded with conditional mean converging to $D^\star$ under the growth-rate-optimal predictable plug-in; by the strong law for martingale differences, $\frac{1}{n}\log W_{\tau+n} \to D^\star$, so $\log W_{\tau+n} = nD^\star(1 + o(1)) + \log W_\tau$ with $\log W_\tau = O(1)$ under the mixture, and crossing $\log(1/\alpha)$ occurs at $n = \log(1/\alpha)/D^\star(1 + o(1))$. For the growth-rate bound, choose $\lambda = \Delta/(\sigma^2 + B'\Delta)$ and use $\log(1+x) \geq x - \frac{x^2}{2}$ with $\mathbb{E}_{\text{post}}[Z - \varepsilon_1] \geq \Delta$ and $\mathbb{E}_{\text{post}}[(Z - \varepsilon_1)^2] \leq \sigma^2 + B'\Delta$ to obtain $D^\star \geq \Delta^2/(2(\sigma^2 + B'\Delta/3))$. The Lorden/Lai bound [22, 23] states that any procedure with false-alarm rate $\leq \alpha$ has worst-case delay $\geq \log(1/\alpha)/D_{\text{KL}}(1+o(1))$; since $D^\star \leq D_{\text{KL}}$, ATLAS attains this order. □

Remark 4 (Reading)**.** Theorem 3 quantifies Figure 1: after a dynamics shift of margin Δ, the trust horizon collapses within $O(\log(1/\alpha)/\Delta^2)$ rounds in the small-margin regime. When the bet saturates at $\lambda_{\max}$ (large margin), $D^\star \propto \Delta$ and the delay scales as $1/\Delta$; Section 5.1 exhibits both regimes.

5. Experiments

Our study is an ascending ladder of realism. We first validate each theorem in a controllable latent simulator where the ground-truth dynamics—and hence the true faithfulness—are known exactly (Section 5.1); this is the only setting in which the guarantees can be checked against truth, since for a real world model "truth" is unobserved. We then move to real image-sequence data under three distinct distribution shifts (Section 5.2) and finally to learned neural world models trained on a GPU (Section 5.3).

Table 1: Theorem 1 in practice: empirical false-revocation rate under the null (400 deployments, $\alpha = 0.05$, per-horizon level $\alpha/H = 0.01$). ATLAS never falsely revokes a faithful horizon; the rates sit far below the guarantees.

horizon h	1	2	5	10	20	any
empirical revocation rate	0.000	0.000	0.000	0.000	0.000	0.000
guarantee (bound)	0.01	0.01	0.01	0.01	0.01	0.05

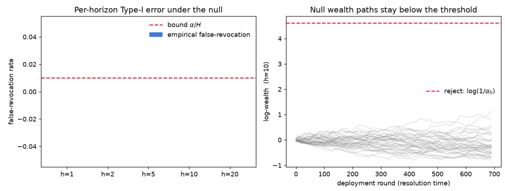
5.1. Ground-truth validation of the three theorems

The environment is a stable linear-Gaussian latent system $z_{t+1} = Az_t + Ba_t + w_t$; the world model uses the true pre-change parameters, so it is exactly faithful until an unmodelled force is injected at a known changepoint. A behavior policy draws exogenous exploration actions, which makes the action-conditioned predictive exactly calibrated under the null (so there is a genuine null to test) while keeping the state stream non-independent and adapted. The score statistic is the calibration excess $Z = \text{mahalanobis}^2/d - 1$, processed on non-overlapping windows.

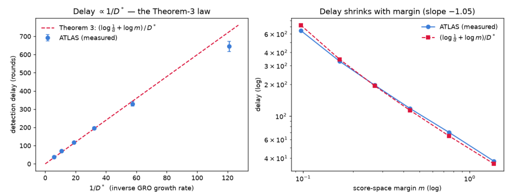
Theorem 1 (no false revocation). Under the null, across 400 independent deployments, the empirical revocation rate is at or below the level, uniformly over all rounds (Table 1, Figure 2). Every horizon is well within its $\alpha/H = 0.01$ budget and the any-horizon rate is far below $\alpha = 0.05$; the e-process is conservative, exactly as Ville's inequality predicts.

Theorem 2 (the frontier as a controller). With a drift injected at round 900, the certified one-step gap $\hat{\delta}_1(t)$ tightens toward the true gap while the model is faithful and rises as soon as the drift opens; the trust horizon $h^\star(t) = \max\{h : C(h, \gamma)\phi(\hat{\delta}_1(t)) \leq \text{budget}\}$ collapses from 38 to 7 within a few rounds, while an offline metric frozen at its pre-deployment value remains at 38 (Figure 1). This is exactly the controller behavior the theorem licenses.

Theorem 3 (detection delay). Sweeping the drift margin and measuring the detection delay over 100 seeds, the measured delay matches the theoretical law $(\log(1/\alpha) + \log m)/D^\star$ to within $\approx 6\%$, with delay $\cdot\, D^\star \approx 6$ nearly constant, confirming delay $\propto 1/D^\star$ (Figure 2, right). In this range $D^\star \propto m$ because the bet saturates at $\lambda_{\max}$, so the delay scales as $1/m$; the $1/m^2$ small-margin regime requires $m \lesssim 0.07$, consistent with Remark 4.



(a) Uniform validity: per-horizon Type-I error under the null (left) and null wealth paths staying below the $\log(1/\alpha_h)$ threshold (right).



(b) Detection delay matches $(\log \frac{1}{\alpha} + \log m)/D^{\star}$ (left) and is linear in $1/D^{\star}$ (right).

Figure 2: Ground-truth validation of Theorems 1 and 3.

5.2. Real-world data

We monitor a learned latent world model—principal-component features with a stabilized linear latent dynamics model, a deliberately lightweight but genuine latent simulator—on real image sequences. The tolerance ε_h is calibrated on held-out in-distribution clips, on the same statistic the test consumes (real-video score excess is heavy-tailed, so a naive window-level calibration is mis-scaled); deployment then runs in-distribution before switching regime at a known round. We study three shifts spanning two datasets and two failure modes.

Moving MNIST [25]: a speed (dynamics) shift. The stream runs at the native frame rate, then switches to double speed—a dynamics change the native-rate model cannot track. The wealth is flat during the native period (valid, no depletion) with $h^{\star} = 8$; after the shift, horizons 1–3 cross the threshold and $h^{\star}$ collapses to 0, while the offline trust horizon stays at 8 (Figure 4, left).

KTH Actions [26]: a gait (dynamics) shift. A world model fit on *walking* video is deployed; the subject then switches to *running* (faster leg dynamics). $h^{\star} = 3$ holds throughout the walking period with no false revocation, then collapses to 0 once running begins (Figure 4, middle; sample frames in Figure 3).

KTH Actions: a domain (appearance) shift. The same action (walking), but the recording condition changes from outdoors to indoors. This subtler shift produces a partial collapse, $h^{\star} : 3 \to 2$ (only the longest horizon is revoked); the graded response—full collapse for strong dynamics shifts, partial for a subtle appearance shift—is itself informative (Figure 4, right).

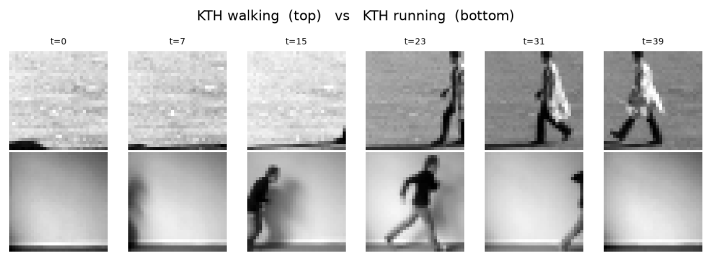
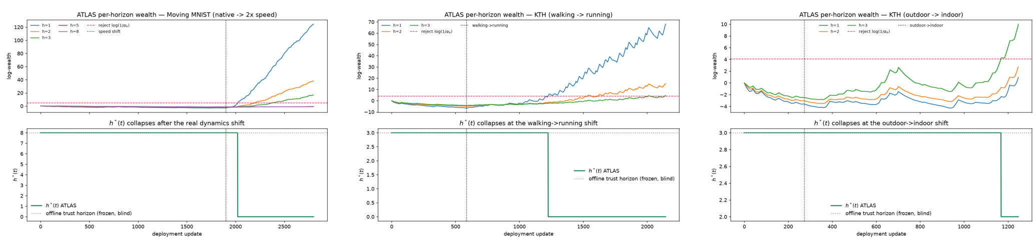


Figure 3: Real human-action video from KTH: walking (top) and running (bottom). ATLAS monitors a walking-trained world model and detects the switch to running.



(a) Moving MNIST (speed). (b) KTH walking→running. (c) KTH outdoor→indoor.

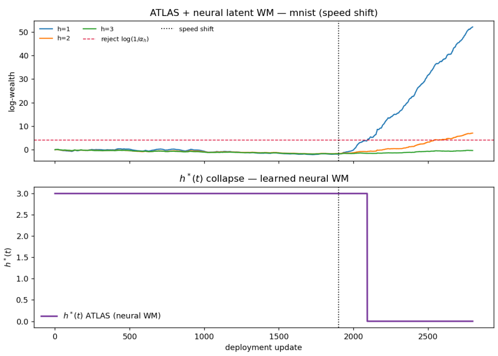
Figure 4: Real-data frontier collapse under three distinct shifts. In every case the per-horizon wealth is flat during the in-distribution period (valid), and $h^\star(t)$ collapses when the shift opens—fully for the two dynamics shifts, partially for the subtle domain shift.

5.3. Learned neural world models

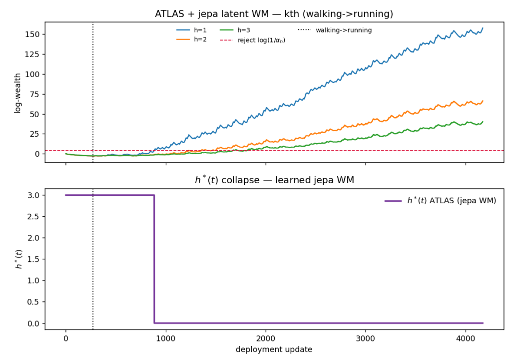
Finally we replace the linear world model with learned neural world models—which expose the same interface, namely a per-horizon predictive distribution scored against the realized outcome—to show that ATLAS's guarantees are indifferent to model class. These experiments were run on a graphics processing unit.

Convolutional-autoencoder world model on Moving MNIST, a latent-imagination model in the Dreamer family. The wealth is flat during the native period; $h^\star$ collapses from 3 to 0 after the double-speed shift (Figure 5, left). ATLAS's guarantees hold behind a genuinely learned model.

JEPA-style world model on KTH walking→running. A frozen pretrained image encoder with a learned latent-dynamics network generalizes across subjects, so the in-distribution walking score excess is low and stable while running is clearly separated. $h^\star = 3$ holds through the entire walking period (a valid null) and collapses to 0 once running begins (Figure 5, right). This experiment isolates a practical lesson: a representation that generalizes across



(a) Convolutional-autoencoder model, Moving MNIST (speed).



(b) JEPA-style model, KTH walking→running.

Figure 5: ATLAS monitoring learned neural world models: a valid in-distribution null (flat wealth) followed by $h^\star(t)$ collapse at the shift, for both a convolutional autoencoder and a JEPA-style model.

the nuisance factors of the deployment (here, subject identity) is what makes the certificate clean on real video, and a stronger world model such as a recurrent state-space model can be substituted without any change to the certification procedure.

Three findings recur across all tiers. First, validity is never violated: during every in-distribution period the wealth is flat and $h^\star$ holds, matching Theorem 1. Second, the frontier collapses at the shift—sharply for dynamics shifts and partially for a subtle appearance shift—always before an offline metric would react, indeed the offline horizon never reacts at all. Third, the behavior is model-agnostic, holding for linear, convolutional-autoencoder, and JEPA-style world models, as the theory requires.

6. Discussion and conclusions

We have introduced ATLAS, which gives a deployed world model something it has never had: a running, anytime-valid certificate of faithfulness. The certificate is a trust horizon $h^\star(t)$ that collapses precisely when reality drifts from imagination, backed by three guarantees we proved in full—a uniform no-false-revocation theorem, an anytime-valid simulation lemma linking certified one-step error to the h-step value gap, and an order-optimal detection-delay bound. The construction is representation- and model-agnostic, requires no exchangeability, and converts a passive monitor into a planning controller by clipping imagination depth to the certified horizon. Across a ground-truth simulator, real human-action and video-benchmark data, and learned neural world models, ATLAS behaves exactly as the theory predicts, while the offline metrics in universal use today stay blind.

Several limitations delineate the scope and point to future work. Our controlled study uses an exogenous behavior policy to obtain an exact null against which Type-I error can be measured; the fully policy-in-the-loop regime is the ultimate target, and while the conditional construction remains valid there for the same reason, quantifying power under closed-loop feedback is natural future work. Non-overlapping windows make long horizons genuinely slower to certify because they receive fewer effective samples; the interleaved variant of Remark 1 mitigates but does not eliminate this intrinsic price of honest multi-step testing. Certificate quality is bounded by world-model quality: a low-capacity or overfit model produces a non-stationary in-distribution null, which can mask or spuriously trigger revocation, and our JEPA experiment makes the remedy concrete—a representation that generalizes across nuisance factors restores a clean certificate. Finally, the tolerances ε_h are modeling choices that define what "faithful" means; we calibrate them to the in-distribution level so that ATLAS certifies departures from the model's own baseline, but a fixed absolute tolerance or a risk-derived budget is equally admissible and changes only the interpretation, not the validity. As world models move from benchmarks into deployed decision loops, we believe a running, valid faithfulness certificate will be not a luxury but a requirement, and ATLAS offers a principled foundation for it.

Data and code availability

The methods and all reported experiments are reproducible from an accompanying open implementation together with the publicly available Moving MNIST and KTH Actions datasets.

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